

Thomas Madden

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Education

University of California, San Diego, B.S. in Mathematics

September 2023 – June 2027

- GPA: 3.89/4.0; Provost Honors all quarters

Research Experience

UC San Diego Mathematics Honors Program

September 2026 - Present

- Acceleration for Randomized Kaczmarz and randomized low-rank approximation.
- Advised by Robert J. Webber

UCLA Computational and Applied Mathematics (CAM) REU

June 2026 – August 2026

- Proved sharp convergence guarantees for Fisher-Rao and Wasserstein-Fisher-Rao gradient flows
- Identified properties of initializations which lead to better performance of algorithms based on these flows for sampling from mixture distributions
- Clarified necessary and sufficient conditions for convergence of the flows
- Advised by Yifan Chen

Scalable Optimization and Control (SOC) Lab at UC San Diego

May 2025 – Present

- Introduced algorithm that generically accelerates the proximal bundle method in convex optimization
- Showed that incorporating iterate-acceptance procedure allows large stepsize selection
- Performed numerical simulations using JAX, CVXPY, and NumPy to illustrate speed-up
- Advised by Yang Zheng

Random matrices and Λ -distributions, Polymath Jr.

June 2025 – September 2025

- Proved exact criterion for independence of classical random variables obtained from Λ -random variables, with applications in random matrix statistics and number theory
- Presented results on behalf of 25+ person group at end-of-project conference
- Advised by Sean Howe

Teaching Experience

Content Tutor, Teaching + Learning Commons at UC San Diego

September 2024 – June 2025

- Tutored undergraduates in all lower division calculus courses, linear algebra, and mathematical reasoning
- Led sessions with 1-3 students from different classes attending simultaneously
- Incorporated abstract motivation for concepts in calculus, along with physical intuition
- Received 100% recommendation rating from students

Course Facilitator, Subject Inc.

February 2024 – September 2024

- Graded assignments and provided individualized feedback for 300+ students in Algebra II and Geometry
- Made 100+ adjustments to curriculums to improve clarity/correctness of solution manuals
- Tutored students as needed to maintain $\sim 90\%$ improvement in mathematical aptitude

Papers

Published:

1. *An accelerated proximal bundle method for convex optimization*
F.-Y. Liao, T. Madden, and Y. Zheng, *L4DC 2026*, Proceedings of Machine Learning Research **331** 2026, 1012–1034. <https://proceedings.mlr.press/v331/liao26a.html>

Preprints:

1. *On the convergence of Fisher-Rao and Wasserstein-Fisher-Rao gradient flows of the KL divergence*
T. Madden and Y. Chen. *In preparation*.
2. *Matrix group Λ -distributions*
M. Bertucci, S. Van Blarcom, V. Glancy, S. Howe, T. Madden, B. Miller, S. Pal, G. Papagrigoriou, N. Raikman, and T. Tran. *Submitted*, 2025. <https://arxiv.org/abs/2608.16796>

Presentations

Talks:

August 12, 2026 "On the convergence of Fisher-Rao and Wasserstein-Fisher-Rao gradient flows of the KL divergence," UCLA Computational and Applied Mathematics (CAM) REU Research Presentation

February 13, 2026 "Acceleration with large gradient steps via the proximal bundle method," UC San Diego Mathematics of Information, Data, and Signals Seminar.

January 7, 2026 "Random matrices and Λ -distributions," AMS Special Session on Polymath Jr. Student Research.

Posters:

May 28, 2026 "Acceleration with large gradient steps via the proximal bundle method," UC San Diego Undergraduate Engineering Research Symposium (UERS).

Relevant Coursework

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| • MATH 140ABC: Real Analysis | • MATH 240A: Real Analysis (<i>i.p.</i>)* |
| • MATH 100ABC: Abstract Algebra | • MATH 286: Stochastic Differential Equations (<i>i.p.</i>)* |
| • MATH 148: Analysis of Partial Differential Equations | • MATH 280ABC: Probability Theory |
| • MATH 180A: Introduction to Probability | • MATH 281A: Mathematical Statistics |
| • MATH 180B: Introduction to Stochastic Processes I | • MATH 216A: Supervised and Unsupervised Learning |
| • MATH 114: Computational Stochastics | • MATH 216B: Deep Learning |
| • MATH 154: Graph Theory | • MATH 216C: High-Dimensional Probability |
| • MATH 120A: Introduction to Complex Analysis | • MATH 294: The Mathematics of Finance |
| • MATH 105: Basic Number Theory <i>*in progress</i>
<i>abbreviated as i.p.</i> | • DSC 291A: Stochastic Optimization |
| | • DSC 291A: Sampling and Diffusion Models |

Activities

Society of Undergraduate Mathematics Students, President January 2025 - Present

- Created Pi Mu Epsilon chapter at UC San Diego to provide scholarship and conference travel funding opportunities to undergraduates engaged in research
- Planned math-related workshops, including professional development and networking opportunities
- Hosted annual integration bee with original and challenging problems

Society of Undergraduate Mathematics Students Reading Group, Lead December 2025 - Present

- Hold graduate-level reading group on "Statistical Optimal Transport" by Sinho Chewi, Jonathan Niles-Weed, and Philippe Rigollet

Additional

Libraries: JAX, CVXPY, NumPy

Languages: English (native) and Spanish (fluent)

Awards: 2024 UCSD Integration Bee Winner; Received highest score school-wide on Integration Bee Qualifying Exam, HSF Scholar